RocketWire

User Guide

Designing and documenting rocket electrical harnesses

Cornell Rocketry Team

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1. Getting started

RocketWire is shared by the whole team through one team account. Before you start working, **log in**, then **Sync** to pull the latest changes. When you are done making changes, press **Push**.

Button	What it does
Help	Opens this user guide.
↻ Sync	Pulls the cloud copy of your projects onto this device. It also runs automatically when you log in and when you open the site. A project with edits you have not pushed is never overwritten silently: Sync asks before discarding them.
Push	Uploads your changes to the cloud so everyone else sees them. On a project page it pushes only that project. On the home page it pushes every project you changed on this device.
Account pill	Shows who is signed in (green dot).
Log out	Signs out. Editing requires being logged in.

How Sync and Push work together

- Your work is always saved on this device first. Nothing reaches teammates until you press **Push**.
- **Push overwrites the cloud copy.** If two people push the same project, whoever pushes last wins. So always **Sync before you start**, and Push soon after you finish.
- A project counts as "changed" only when its content changed. Panning and zooming never count. If nothing changed, Push tells you "Nothing to push".
- Delete actions are remembered, so a deleted system or wire does not come back after a Sync.

> **Offline login:** if you used the offline fallback login instead of your email account, Push is unavailable. Log out and log in with the team email account.

> **Recommended routine:** Log in → Sync → make your edits → Push → tell the team.

2. The home page: projects and folders

The home page lists your projects and folders. Use the search box to filter by name.

Control	What it does
+ New Project	Creates a project. You can choose which folder it goes in.
+ Folder	Creates a folder. Folders can be nested inside other folders.
↻ Sync	Pulls the latest projects from the cloud.
Push	Pushes every project you changed.

Project menu

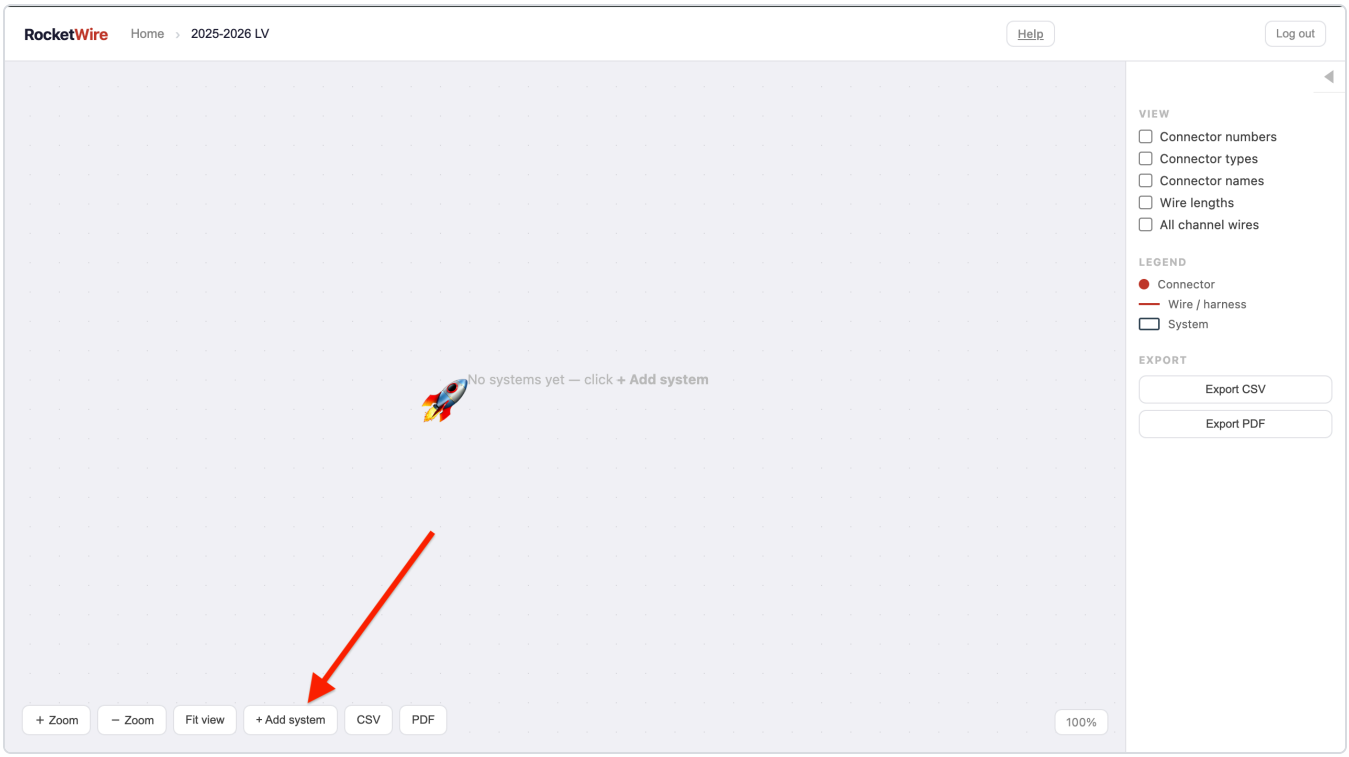
Right-click a project to open its menu:

- **Open** the project.
- **Rename** it.
- **Copy** it, for example to start next year's vehicle from this year's.
- **Move to…** a different folder.
- **Delete project**: asks to confirm, and also removes it from the cloud for everyone.

> Folders have a similar right-click menu for renaming, moving and deleting.

3. Working on the canvas

Opening a project shows the canvas, where systems are boxes, connectors are red dots on a box's edge, and wires join connectors.



An empty project. Tools along the bottom, view options on the right.

Moving around

Action	How
Zoom	Mouse wheel / trackpad scroll, or the `+ Zoom` and `− Zoom` buttons. The percentage shows at the bottom right.
Pan	Hold `Alt` (Option) and drag, or drag with the middle mouse button.
Fit everything on screen	`Fit view`
Move a system	Drag its box.
Slide a connector along its edge	Drag its red dot. Swapping order along an edge helps untangle crossed wires.
Reshape a wire	Drag the small dot at the wire's midpoint. Double-click that dot to reset the wire's shape.
Open a connector	Double-click it, or right-click → View / edit.

The View panel (right side)

- **Connector numbers / types / names**: show labels next to each connector dot.
- **Wire lengths**: show the length on each wire (only for wires where a length was set).
- **All channel wires**: draw every named channel as its own colored line, instead of one red bundle.

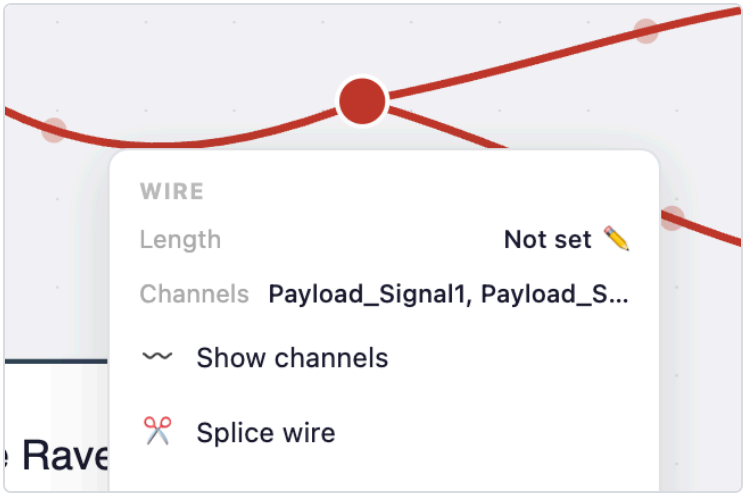
Use the ◀ tab to collapse the panel.

Bundled nets (bus notation)

When a connector has several pins with the *same* channel name, "All channel wires" draws them as **one line** and marks it with the number of wires, for example **X2**. A connector with 12V, 12V, 24V, 24V, GND shows three lines: 12V **X2**, 24V **X2** and GND.

- Channels with no name are not drawn.
- If the same net has different colors on different pins, the bundled line uses the first one's color.

Wire right-click menu



Right-click a wire to set its length, show or hide channels, splice it, or delete it.

- **Length**: click the pencil to set it.
- **Show channels / Hide channels**: overrides the global checkbox for just this wire.
- **Splice wire**, **Edit connector**, **Delete wire**.

Connector right-click menu

View / edit, Convert to splice (or Edit splice), Copy, **Pin to edge** (Auto, Right, Left, Top, Bottom) and Delete connector.

System right-click menu

Enter subsystem view, Rename system, Add connector, Paste connector (appears after you copy one) and Delete system.

4. Undo and redo

Made a mistake? RocketWire keeps up to 100 steps of history per project while it is open.

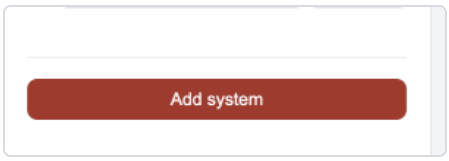
Action	Mac	Windows / Linux
Undo	⌘ + Z	Ctrl + Z
Redo	⌘ + Shift + Z	Ctrl + Shift + Z or Ctrl + Y

It undoes things like moving or deleting systems, connectors and wires, renames, saving a connector, and creating a splice. Undoing a delete brings the item back for good, and it will not be removed again by a later Sync.

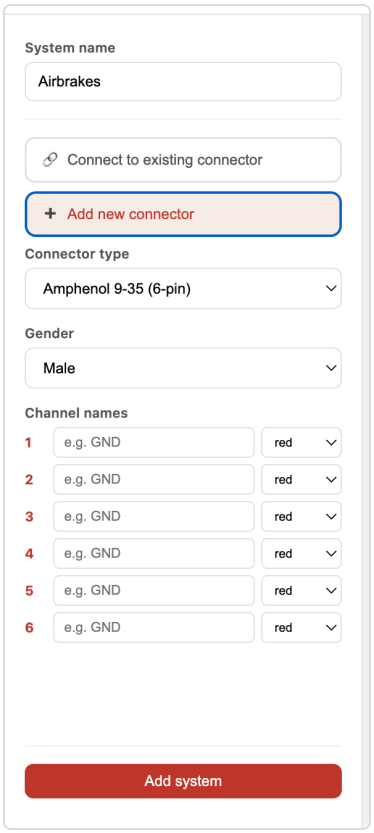
> **Limits:**
>
> - Undo works on the canvas only, not while editing a connector or splice page.
> - Inside a text box, the shortcut undoes your typing instead.
> - History starts fresh each time you open a project, so you cannot undo past a reload.
> - Undo changes your local copy. Press Push to share the result.
> - Panning and zooming are not undo steps.

5. Adding a system

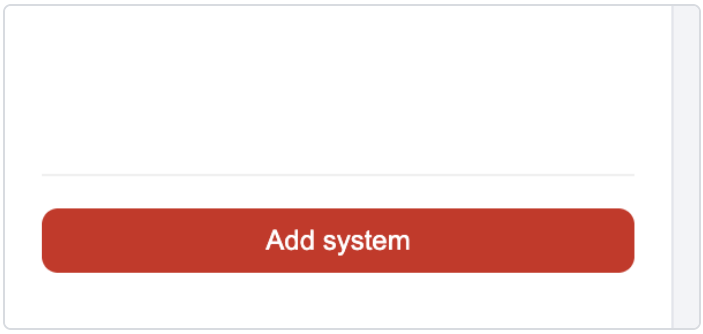
1. Click **+ Add system** in the lower left of the canvas.



2. Type the system name. Choose to connect to an existing connector or add a new connector, then specify the details for that connector (type, gender, channel names and colors).

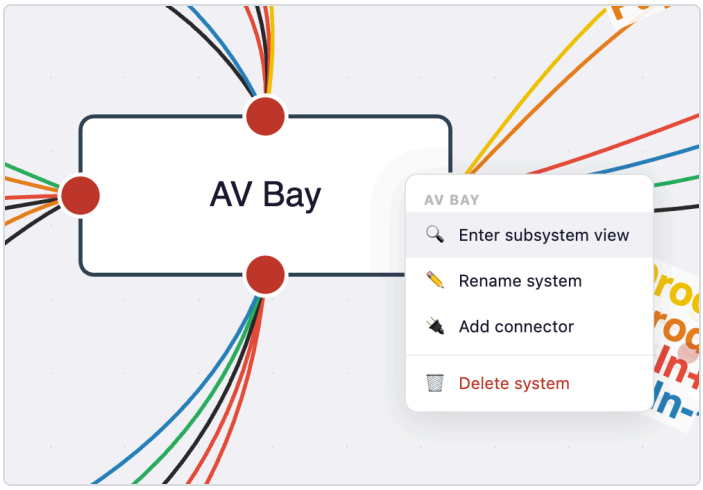


3. Click **Add system** when finished.

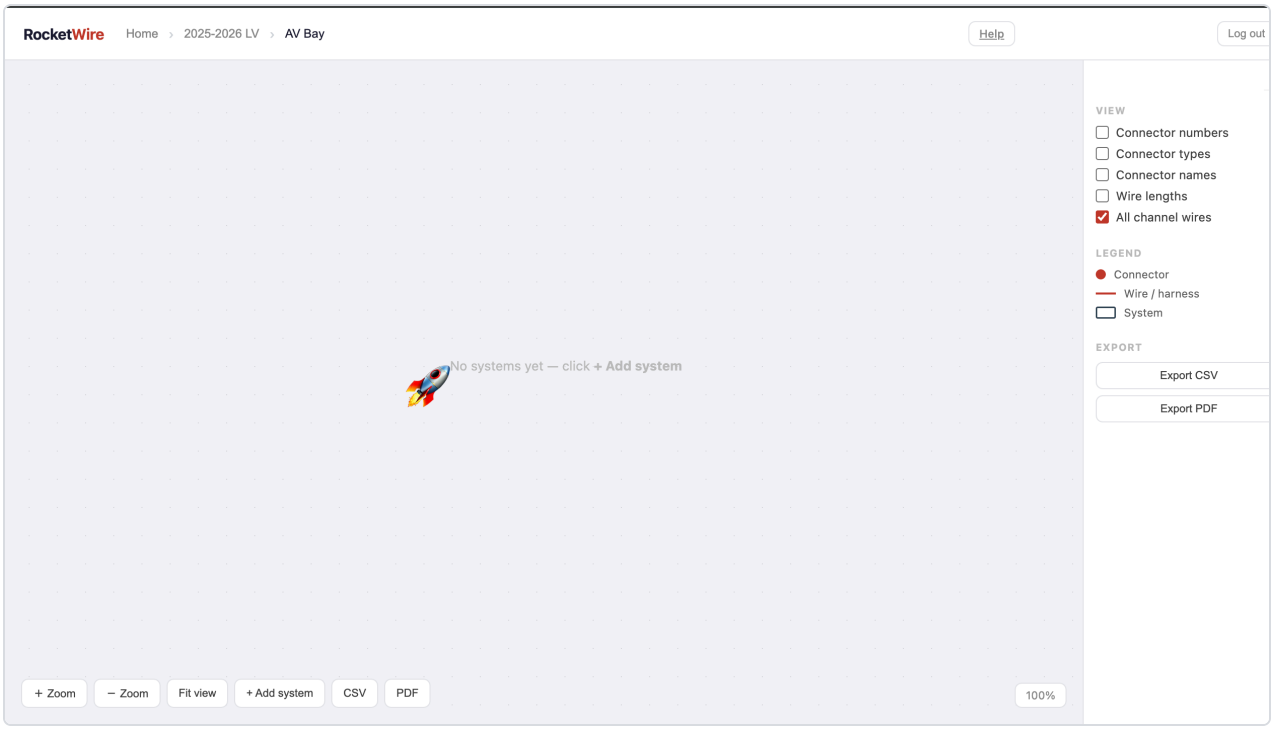


6. Creating a subsystem

1. Right-click a system, then choose **Enter subsystem view**.

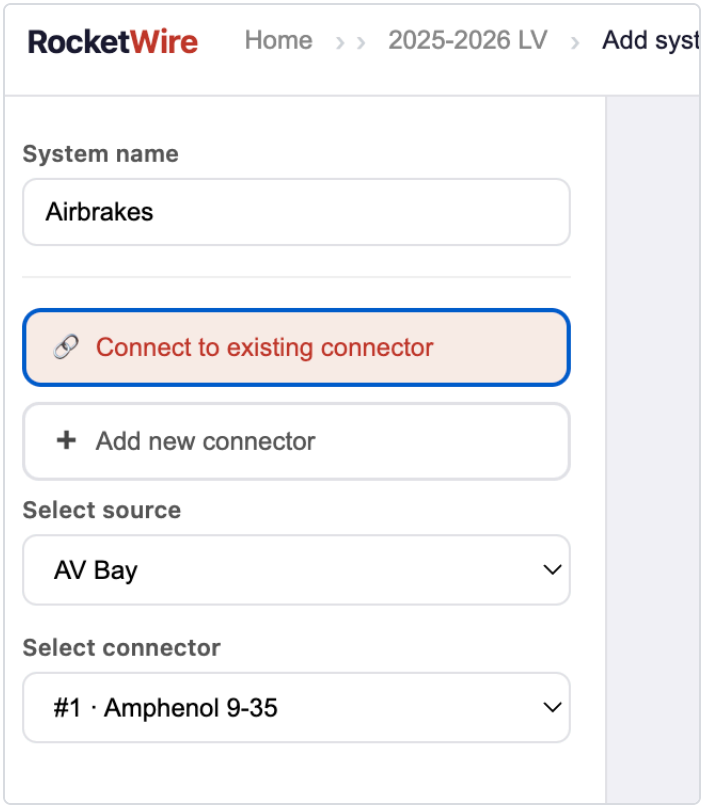


2 Add systems to populate your subsystem. The breadcrumb at the top shows where you are, and you can click it to go back up.



7. Connecting a system to another

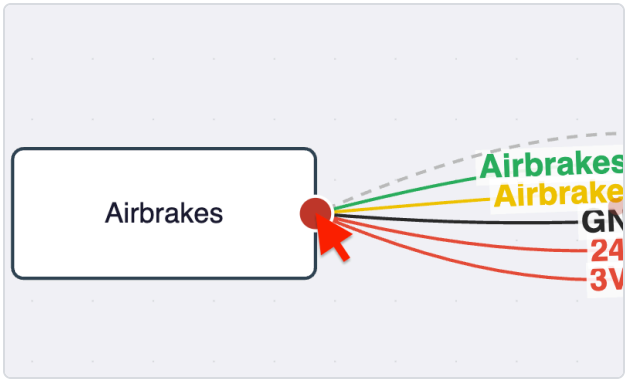
1. Add a system and click **Connect to existing connector**. Choose the system and the connector you want to connect to.

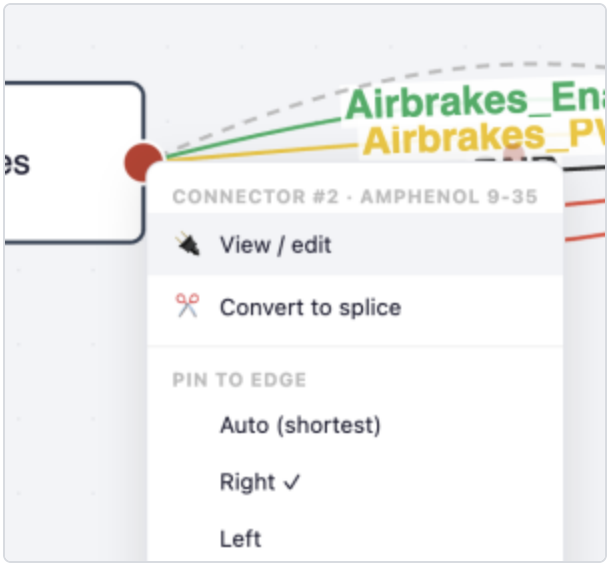


2. Click **Add system** when done.

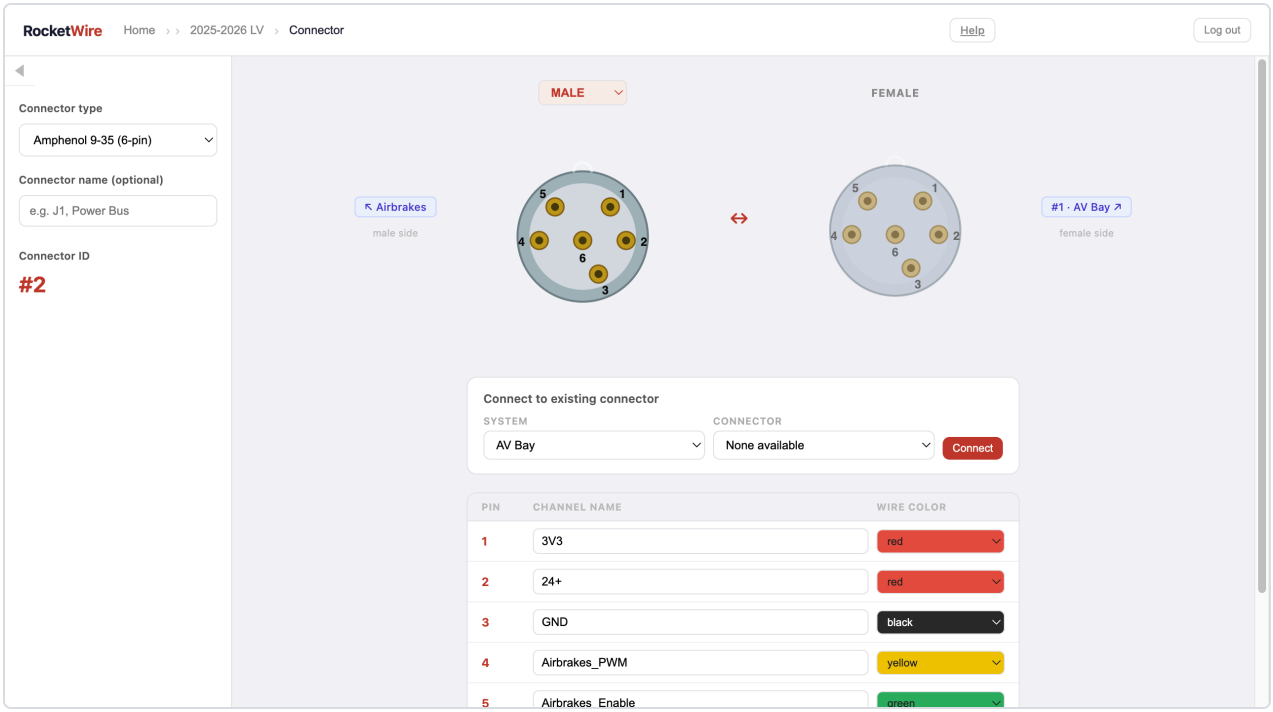
8. Editing a connector

1. Double-click a connector, or right-click it and choose **View / edit**.

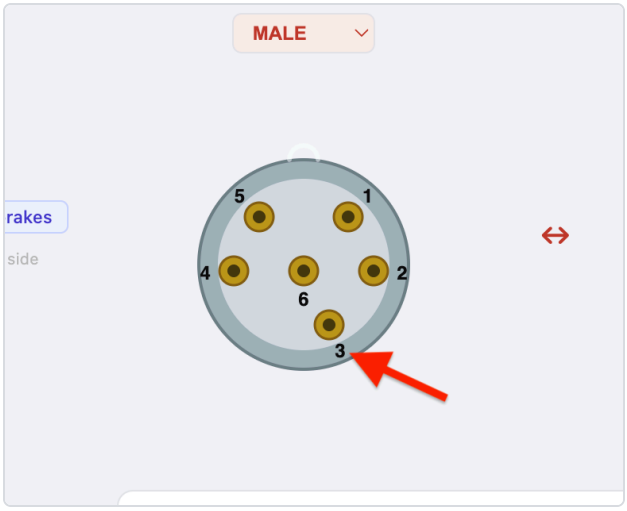


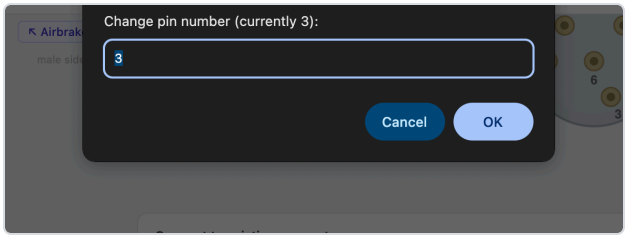


2. In the editor you can change the connector type, give the connector a name (for easy matching later), change the gender, and edit the channel names and colors.

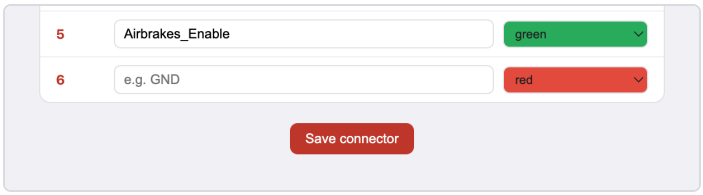


3 If a pin number is incorrect, click the pin number in the diagram. A window pops up to change the numbering.





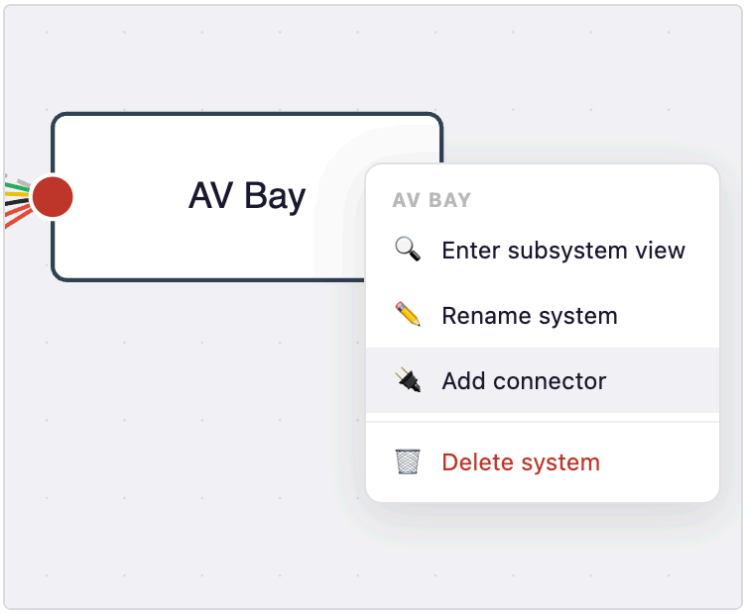
4 Click **Save connector** when done.



9. Adding, copying and pasting connectors

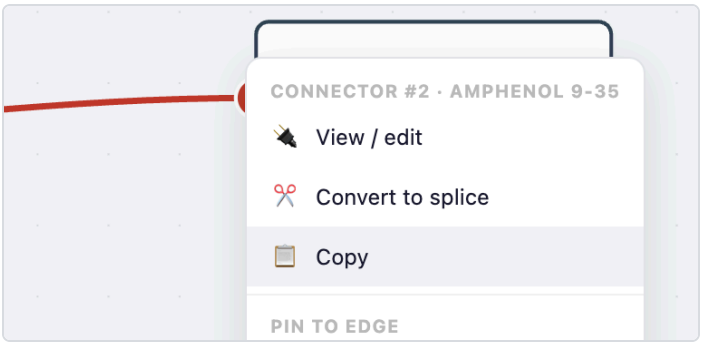
Adding a connector to a system

Right-click a system and click **Add connector**. This opens the same connector editor.

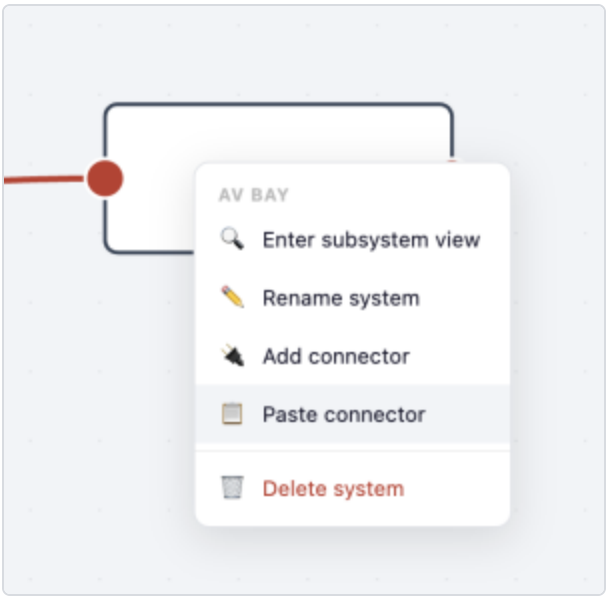


Copying a connector

1. Right-click the connector you want to copy and click **Copy**.



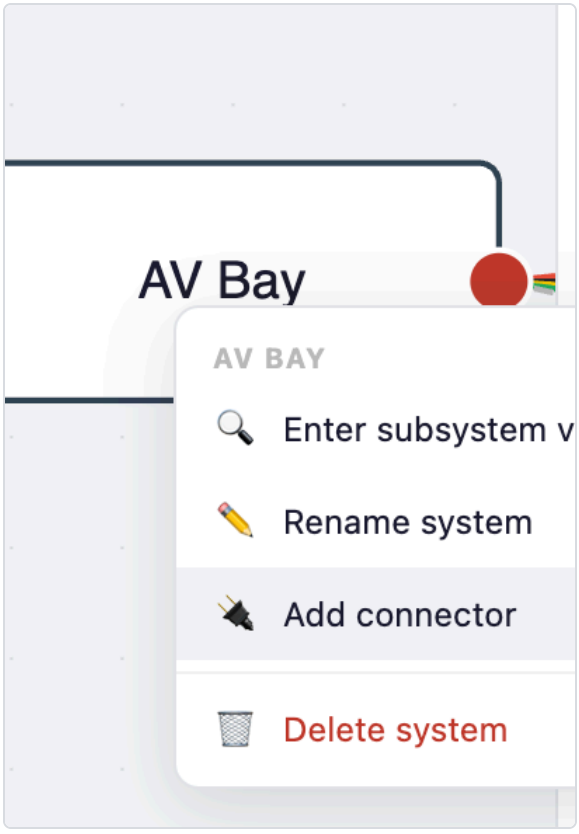
2 Right-click a system and click **Paste connector**.

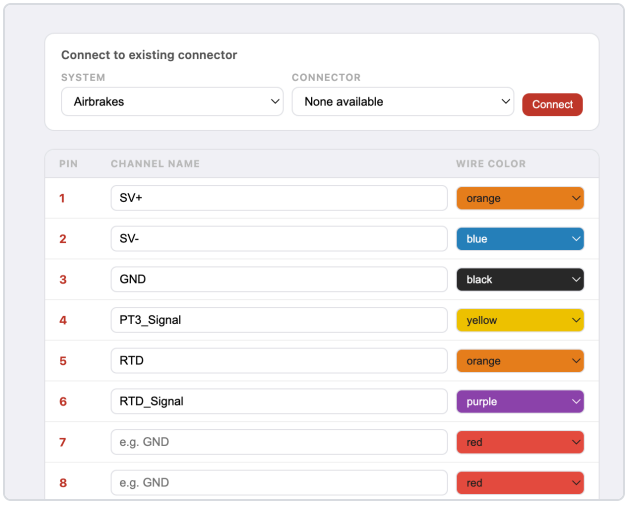


10. Creating a spliced connector

Use a splice when one parent connector fans out to several children. For example, an AV Bay connector that feeds SV, PT3 and RTD.

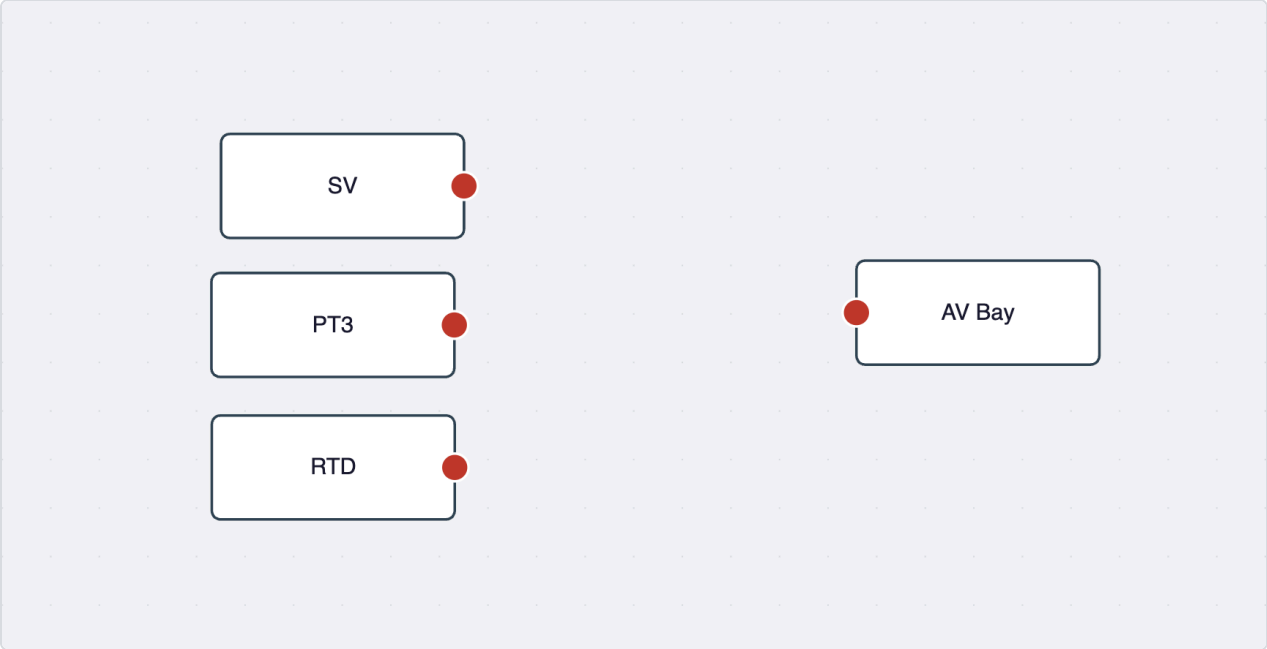
1. Right-click **Add connector** on the parent system (for example the AV Bay). Specify all channel names and colors. Leave unused channels blank. Save the connector.



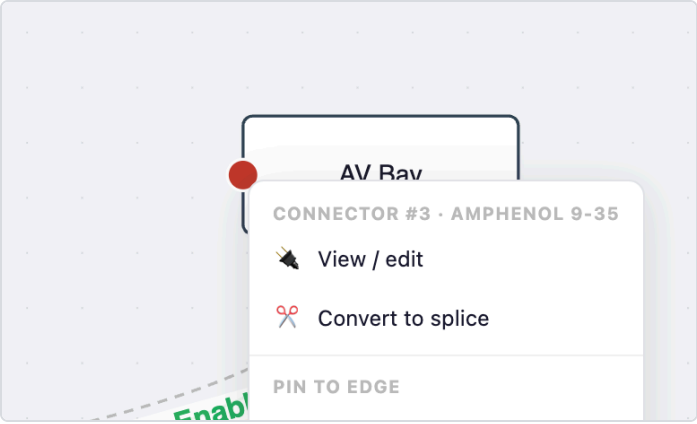


2 Add your child systems and the connectors you are using for them. Specify channel names. There is **no need to specify the color**, since you already did in the parent connector.

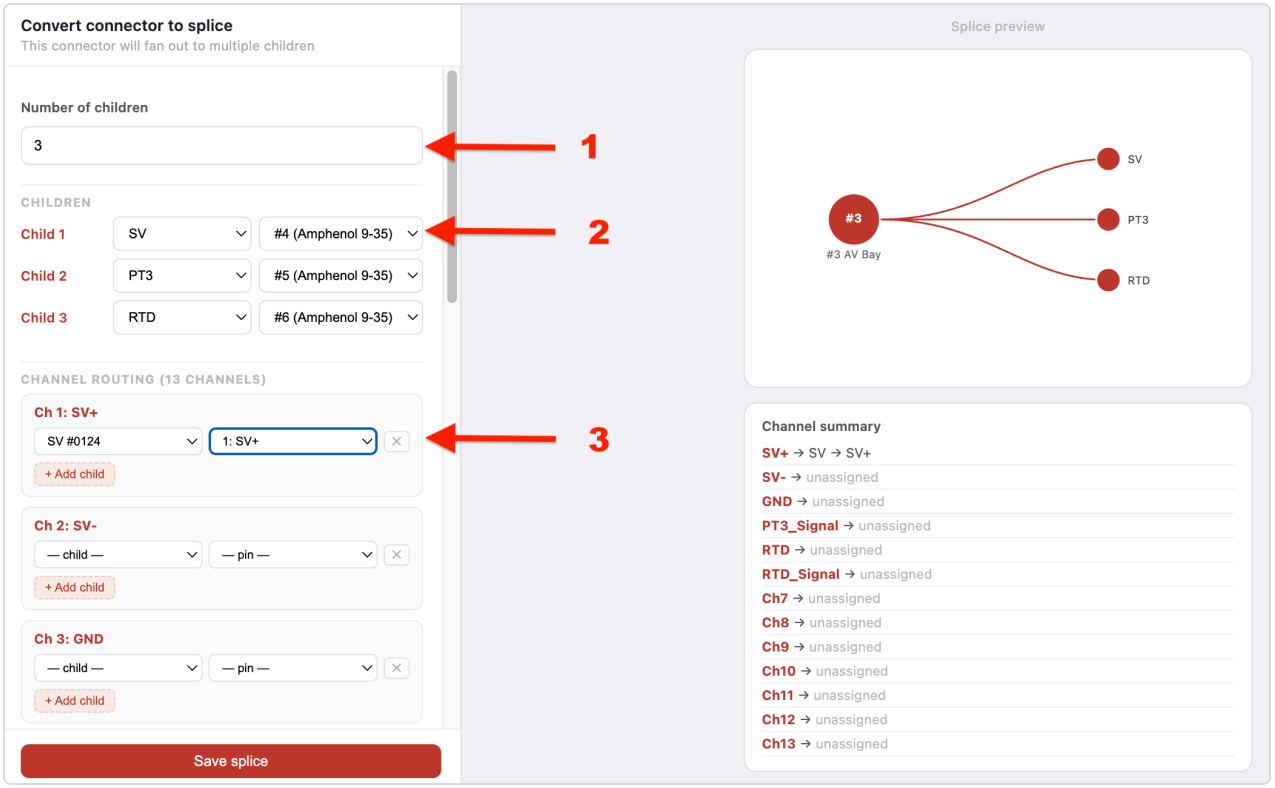
> Do **not** click "Connect to existing connector".



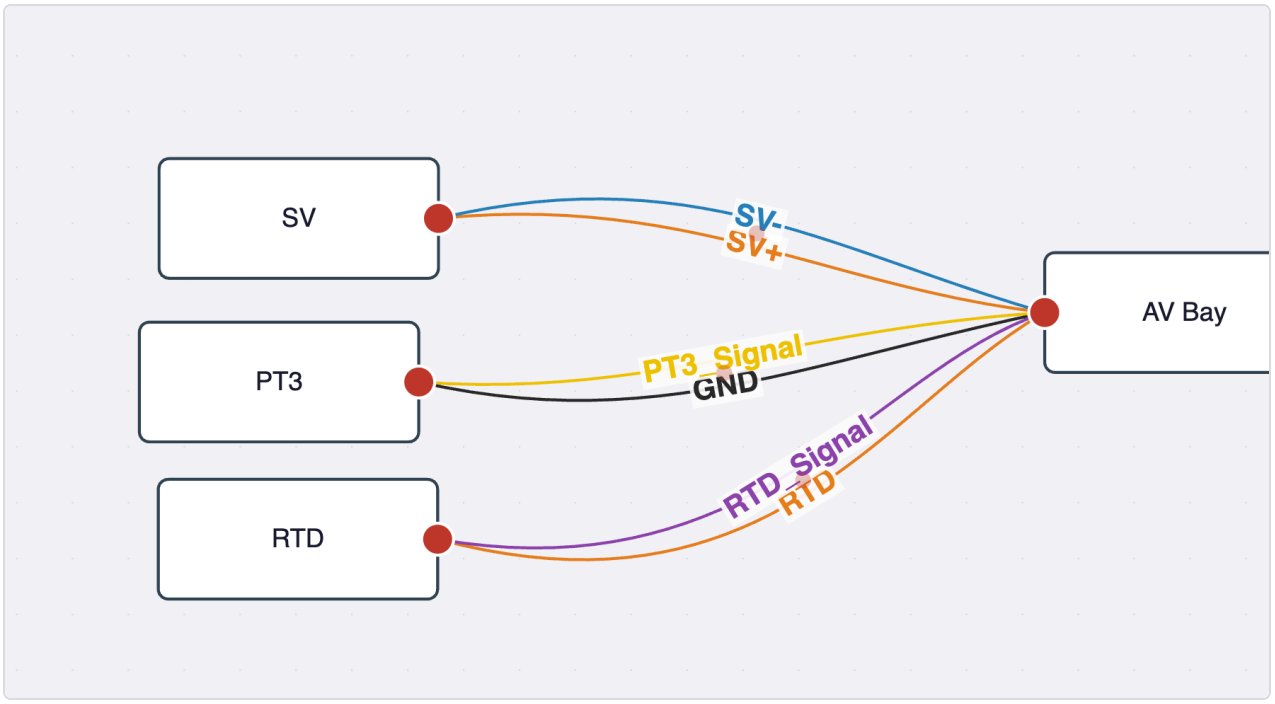
3 Right-click your parent connector, then choose **Convert to splice**.



4 Specify the number of children, what they are, and which connector on each you want to connect to. Then complete the channel routing: for each parent channel, pick the child connector and its channel.

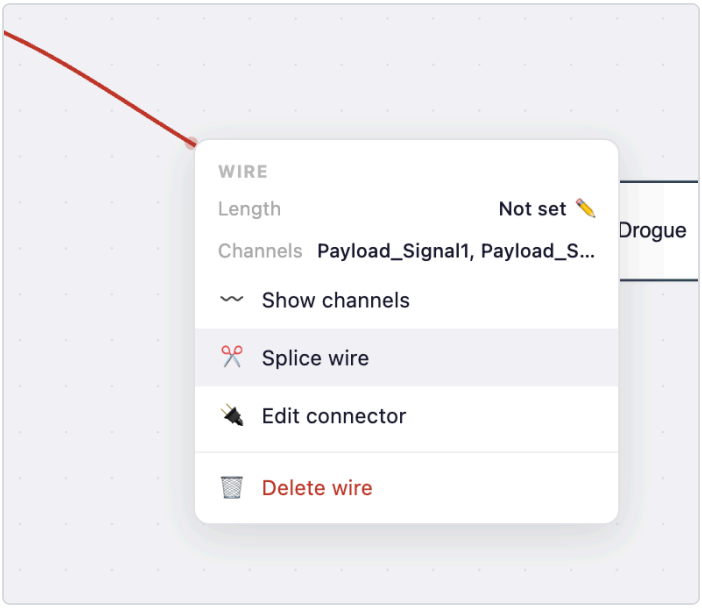


5 Remember to click **Save splice**! Example result:

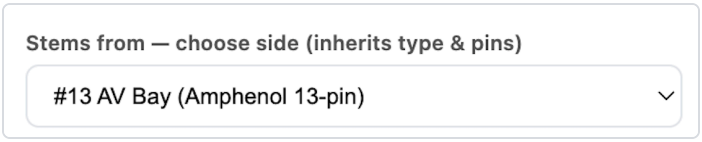


11. Creating a spliced wire

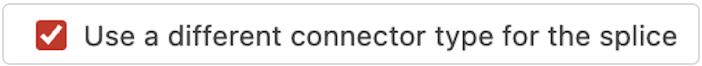
1. Right-click the wire you want to splice, then choose **Splice wire**.



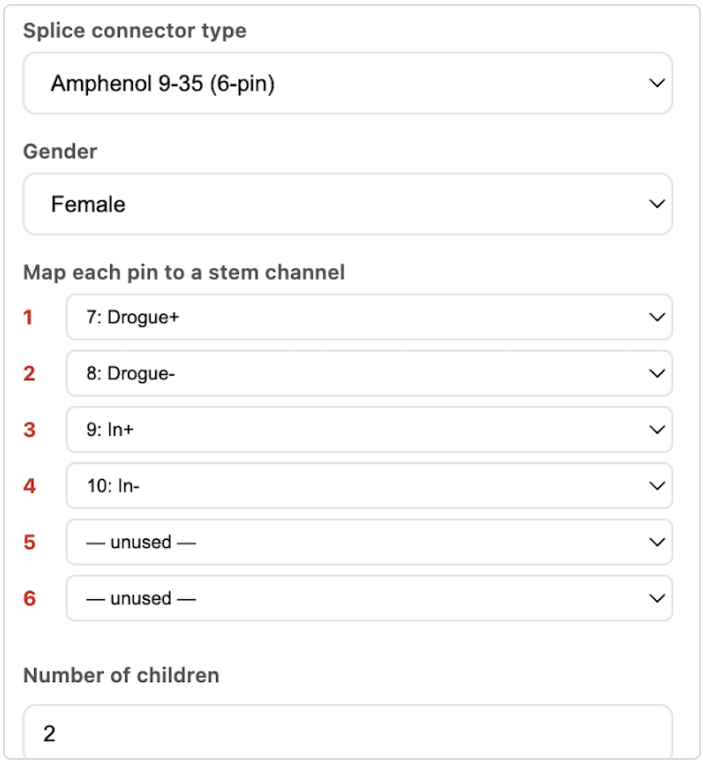
2. Choose where it stems from. This inherits all the channel names (it does not automatically create the channels).



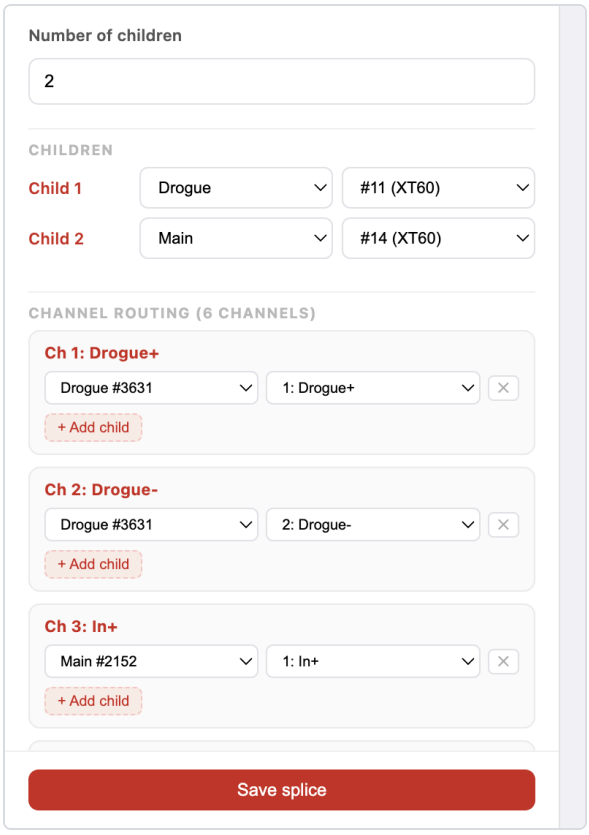
3. You can change the connector type by checking **Use a different connector type for the splice**. If it is unchecked, the new connector is the same as the one it stems from.

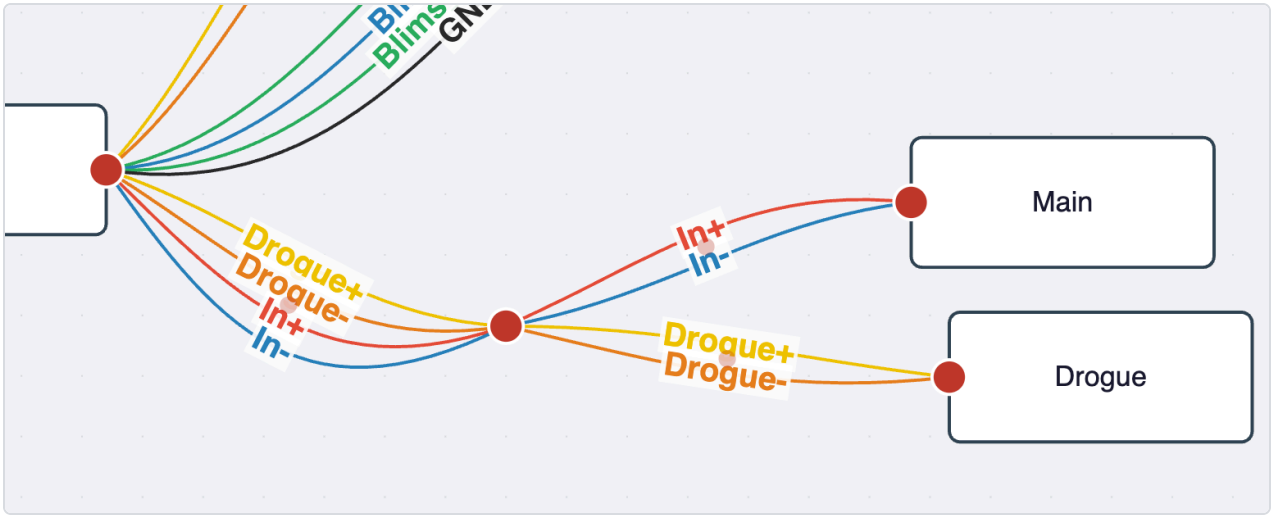


4 Choose the gender, map each pin to a stem channel, and specify the number of children.



5 Map each channel to a child connector, then **Save splice**.





12. Exporting your work

Use the **CSV** and **PDF** buttons on the canvas toolbar, or **Export CSV** / **Export PDF** in the right panel.

- **CSV** downloads a spreadsheet named after your project (for example *My_Project_wiring.csv*) with one row per pin: system, connector number, splice, type, pin count, pin, channel and color.

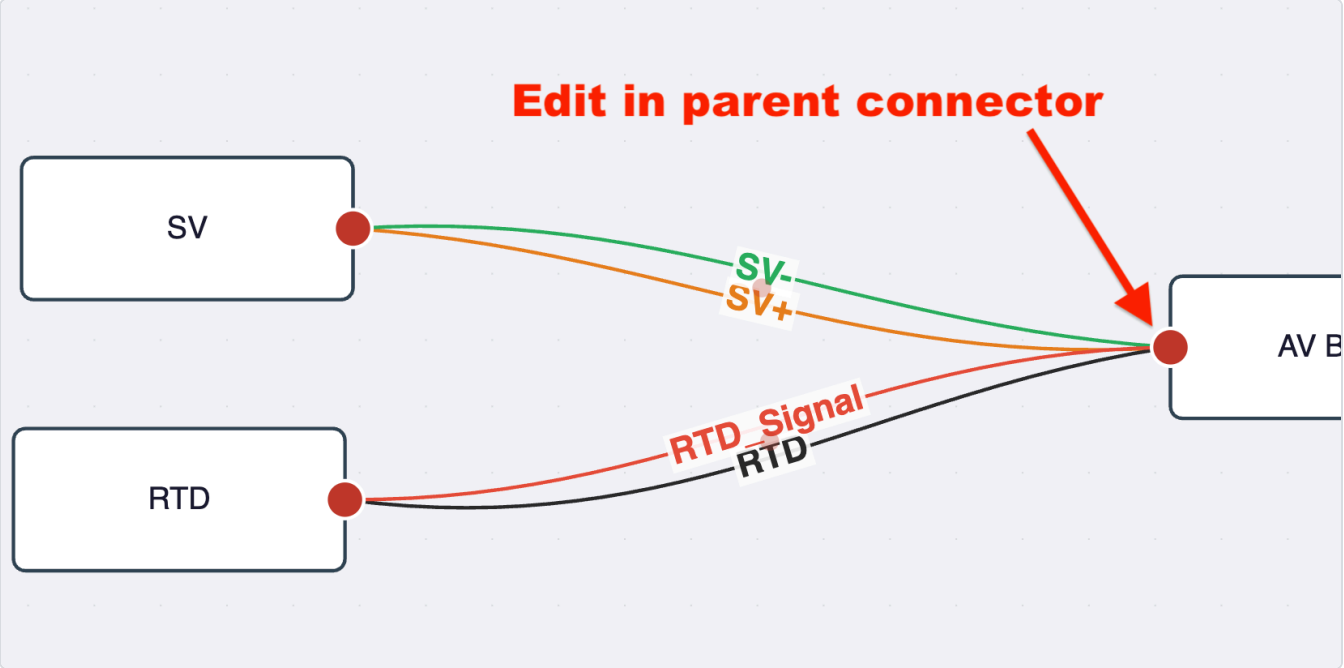
- **PDF** opens a printable wiring report in a new tab. Choose "Save as PDF" in the print dialog.

> Exports cover the level you are viewing. If you are inside a subsystem, you get that subsystem's connectors.

13. Troubleshooting and quick reference

Why can't I edit the channel name of a child connector?

You can only edit the channel name on the **parent spliced connector**. It then updates the child connector automatically.



Why can't I see the wire length?

The length shows only once it is set. Right-click the wire, then set the length. Also make sure **Wire lengths** is ticked in the View panel.

My teammate can't see my changes.

You need to press **Push**. They then press **Sync** (or reopen the site).

Push says "Nothing to push".

There are no content changes since your last Push or Sync. Panning and zooming do not count.

Sync asks before replacing my project.

You have edits that were never pushed. Cancel and press Push first to keep them, or accept to discard them and take the cloud version.

Push is blocked for one project.

The message names the project. This usually means its cloud copy belongs to a different account. Make sure everyone logs in to the same team account.

Quick reference

I want to…	Do this
Get the latest work	↻ Sync
Share my work	Push
Undo / redo	`⌘/Ctrl` + `Z` / `⌘/Ctrl` + `Shift` + `Z`
Pan	`Alt` + drag
Zoom	Scroll wheel
Edit a connector	Double-click it
See every channel	View → All channel wires
Straighten a wire	Double-click its midpoint dot